

Alex Dytso

Curriculum Vitae

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Education

Current	Ph.D., Electrical Engineering	University of Illinois at Chicago (UIC),
2014	Master of Science, Electrical Engineering	University of Illinois at Chicago (UIC),
2011	Bachelor of Science, Electrical Engineering	University of Illinois at Chicago (UIC).

Ph.D. Thesis

Title: Discrete Inputs in Gaussian Interference Channels: Performance Analysis and Approximate Optimality

Advisors: Daniela Tuninetti and Natasha Devroye

Honors and Awards

- 2011-12 Wexler Award of Excellence. Awarded to promising graduate students.
- 2011 International Engineering Consortium's William L. Everitt Student Award of Excellence. Honors outstanding seniors who have demonstrated an interest in the communications field.
- 2010 UIC Engineering Alumni Association Scholarship. Awarded to each year to two students who demonstrate high academic achievements and strong desire to enhance their education by demonstrating leadership skills.
- 2010 Irene Jones Scholarship. Support the next generation of engineers - the men and women who will change the world.

Publications

Peer Reviewed Journal Articles

1. A. Dytso, S. Rini, N. Devroye, and D. Tuninetti, "On the capacity region of the two-user interference channel with a cognitive relay", *IEEE Transactions on Wireless Communications*, vol. **13**, no. (12), 6824–6838, Dec. 2014.
2. A. Dytso, D. Tuninetti, and N. Devroye, "Interference as Noise: Friend or Foe?" *To appear in IEEE Transactions on Information Theory*, <http://arxiv.org/abs/1506.02597>, Jun. 2015.
3. —, "On the two-user interference channel with lack of knowledge of the interference codebook at one receiver", *IEEE Transactions on Information Theory*, vol. **61**, no. (3), 1257–1276, Mar. 2015.
4. A. Dytso, R. Bustin, D. Tuninetti, N. Devroye, H. Poor, and S. Shamai(Shitz), "New bounds on MMSE and applications to communication with the disturbance constraint, *To be submitted to IEEE Transactions on Information Theory*", Mar. 2016.

Invited Conference Papers

5. A. Dytso, D. Tuninetti, and N. Devroye, "On discrete alphabets for the two-user Gaussian interference channel with one receiver lacking knowledge of the interfering codebook", *Proceedings IEEE International Symposium on Information and its Applications (ITA)*, 1–8, Feb. 2014.
6. —, "Nearly optimal non-Gaussian codes for the Gaussian interference channel", *49th Asilomar Conference on Signals, Systems and Computers*, Oct. 2015.
7. A. Dytso, R. Bustin, D. Tuninetti, N. Devroye, H. Poor, and S. Shamai(Shitz), "On communications through a Gaussian noise channel with an MMSE disturbance constraint", *Proceedings IEEE International Symposium on Information and its Applications (ITA)*, 1–8, Feb. 2016.

Peer Reviewed Conference Papers

8. A. Dytso, N. Devroye, and D. Tuninetti, "On the capacity of the symmetric interference channel with a cognitive relay at high snr", *Proceedings IEEE International Conference on Communications (ICC)*, 2350–2354, Jun. 2012.
9. —, "The sum-capacity of the symmetric linear deterministic complete k-user z-interference channel", *50th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, 1232–1237, Oct. 2012.
10. A. Dytso, D. Tuninetti, and N. Devroye, "On the capacity of interference channels with partial codebook knowledge", *Proceedings IEEE International Symposium on Information Theory (ISIT)*, 2039–2043, 2013.
11. A. Dytso, N. Devroye, and D. Tuninetti, "On Gaussian interference channels with mixed Gaussian and discrete inputs", *Proceedings IEEE International Symposium on Information Theory (ISIT)*, 261–265, Jun. 2014.
12. A. Dytso, D. Tuninetti, and N. Devroye, "i.i.d. mixed inputs and treating interference as noise are gDof optimal for the symmetric Gaussian two-user interference channel", *Proceedings IEEE International Symposium on Information Theory (ISIT)*, 1716–1720, Jun. 2015.
13. —, "The Gaussian interference channel with lack of codebook knowledge at one receiver symmetric capacity to within a gap with a PAM input", *Proceedings IEEE Information Theory Workshop (ITW)*, 1–5, Apr. 2015.
14. A. Dytso, R. Bustin, D. Tuninetti, N. Devroye, H. Poor, and S. Shamai(Shitz), "On the applications of the minimum mean p-th error (MMPE) to information theoretic quantities", *Submitted to the IEEE Information Theory Workshop (ITW)*, 2016.
15. —, "On the minimum mean p-th error in Gaussian noise channels and its applications", *Submitted to the IEEE International Symposium on Information Theory (ISIT)*, 2016.

Poster Presentations, Talks and Lectures

Poster Presentations

16. A. Dytso, N. Devroye, and D. Tuninetti, "The sum-capacity of the symmetric linear deterministic complete k-user z-interference channel", *IEEE International Symposium on Information Theory (ISIT)*, Jul. 2012.
17. —, "The sum-capacity of the symmetric linear deterministic complete k-user z-interference channel", *North American School of Information Theory*, Jun. 2012.
18. A. Dytso and H. Safavi, "Error correction codes for TDRS satellites", *NASA Goddard Space Flight Center*, Aug. 2014.
19. A. Dytso, N. Devroye, and D. Tuninetti, "Discrete inputs in Gaussian interference networks: Performance analysis and approximate optimality", *IEEE International Symposium on Information and its Applications (ITA)*, Feb. 2015.

Talks and Lectures

20. "The sum-capacity of the symmetric linear deterministic complete k-user z-interference channel", *North American School of Information Theory*, Jun. 2012.
21. "On the capacity of interference channels with partial codebook knowledge", *IEEE International Symposium on Information Theory (ISIT)*, Jun. 2013.
22. "On Gaussian interference channels with mixed Gaussian and discrete inputs", *IEEE International Symposium on Information Theory (ISIT)*, Jun. 2014.
23. "On the optimality of treating interference as noise in competitive scenarios", *University of Maryland College Park, Host: Sennur Ulukus*, Jul. 2014.
24. "i.i.d. mixed inputs and treating interference as noise are gDof optimal for the symmetric Gaussian two-user interference channel", *IEEE International Symposium on Information Theory (ISIT)*, Jun. 2015.

25. “The Gaussian interference channel with lack of codebook knowledge at one receiver symmetric capacity to within a gap with a PAM input”, *IEEE Information Theory Workshop (ITW)*, Apr. 2015.

Professional Training

Seminars and Workshops

26. “North american school of information theory”, *Seminars on Network Information Theory*, Jun. 2012.
27. “Concentration of measure with applications in information theory, communications and coding”, *IEEE International Symposium on Information Theory (ISIT)*, Jun. 2015.

Professional Affiliations

2014	NASA, Goddard Space Flight Center, Maryland, USA	Summer Internship. Error Correction Codes for TDRS satellites.
2011	China Solar & Clean Energy Solutions Inc. Beijing, China	Summer Internship. Assistant to the main engineer. Performed efficiency of deployment calculations on solar water heaters and solar panels. Developed International Confidence, International Sensitivity and Awareness.
2007-11	Mathematics, Statistics, and Computer Science Tutoring Center. UIC	Math Tutor. Assisted students with understanding and solving low and high-level math problems. Formed study groups.
2009	DRO Electric Co.	Repairing Technician. Setting up, installing and repairing air conditioning units.
2008	SH Electric Inc.	Installation Technician. Installation security cameras, speaker systems, fire alarms, light fixtures, electrical wiring etc.

Teaching Experience

Term	Topic	Courses
Fall-Spring 2014	Random Processes	Wireless Communications
Fall 2015	Capacity of the Continuous Time Channels	Information Theory
Fall 2015	Parallel Channels and Water Filling	Information Theory

Mentoring

I have been a mentor for Naveed Naimipour as a part of his undergraduate research study. Naveed has recently joined our group as a Ph.D. student.

Public Services and Volunteering

2015	Information Theory Society Chicago Chapter	One of the chapter organizers. Did the paper-work, helped to collect petition signatures.
2014-16	Future City Competition	Judge from local IEEE section.
2012-14	Student Information Theory Society	Member. Organized student events at: Allerton 2012: Pictionary Allerton 2012: Pictionary ISIT 2014: Helped organized meeting between Shannon award and students.
2011-12	Local Community Tutoring Center St. Volodymyr Church, Chicago.	Organizer, Tutor.
2009	UIC Flourish of Chicago (an open source conference)	Volunteer.

Related Course Work

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| • Wireless Wireless Communications | • Information Theory |
| • Digital Communications | • Network Information Theory |
| • Detection and Estimation Theory | • Computer Algorithms |
| • Digital Signal Processing | • Measure Theory and Real Analysis |
| • Adaptive Digital Filter Theory | • Complex Analysis |
| • Random Processes Theory | • Functional Analysis |

Projects

Present	MATLAB Projects.	Full list of MATLAB projects can be viewed on MatLabCentral.com. Projects include: Channel Equalizers, scalar Kalman Filter, 3-D Extended Kalman Filter, Combinatorial functions.
2012	Beowulf computer cluster	Using high computing capabilities of computer clusters to simulate and approximate Network Information Problems.
2010-11	Senior Design Project "Watch & White Board"	Using Chronos eZ-430 watch to translate white board notes into the digital format.
2010	RF-Design	Build Fully Operational superheterodyne receiver.
2009	Micro-Controller Project	All designs were performed on the Motorola 68HC11 micro-controller. Projects Included: Stop-Watch/Clock, Count-Down/Up timers, Four-Function Calculator, Electronic Flashcards, Voltmeter.

Computer Skills

- MATAB
- C++
- Java
- Assembly Languages (MASM and Mini Ide)
- VHDL
- LaTeX
- Jemdoc
- LT-Spice, P-Spice
- Office Suits
- OmniGrafle

Membership

- IEEE (Graduate Student Member)
- IEEE Information Theory Society
- IEEE Information Theory Society Chicago Chapter

Languages

- English
- Ukrainian
- Russian

Reference

Reference are available upon request.